

Non-Calculator

On my honor, before God, I have neither given, nor received any aid on this quiz.

Signature _____

You have 30 minutes.

SHOW ALL WORK FOR FULL CREDIT!

1. Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. **(2 pts)**
(More than one of these descriptions may apply.)

a. $-\sqrt{-49}$

b. $6 - 2i$

2. Find the product. Write your answer in standard form. **(2 pts)**

$\sqrt{-8} \cdot \sqrt{-18}$

3. Simplify each power of i . **(2 pts)**

a. i^{47}

b. $1 / i^{14}$

4. Find the sum or difference. Write your answer in standard form.

(2 pts)

$$(6 - 5i) - (-2 + 3i) + (4 - i)$$

5. Find the product. Write your answer in standard form.

(2 pts)

$$(3 - 4i)(-2 + 5i)$$

6. Find the quotient. Write your answer in standard form.

(3 pts)

$$(4 + 7i) / (2 - 3i)$$

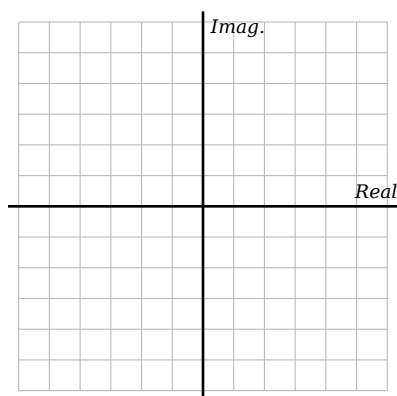
7. Solve the equation over the set of complex numbers.

(4 pts)

$$2x^2 + 3 = -4x$$

8. Find the sum of $5 - 2i$ and $-3 + 6i$. Write your answer in standard form.
Then graph both complex numbers and their resultant.

(3 pts)



9. Write the complex number in rectangular form.

(3 pts)

$$8(\cos 300^\circ + i \sin 300^\circ)$$

10. Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$.

(4 pts)

a. $-3 - 3i$

b. -5

11. The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... If the absolute value of any result exceeds 2, z is **not** in the Julia set. Otherwise it is.

(3 pts)

Is $z = i$ in the Julia set? Show work. YES / NO

Answer Key • Practice Quiz A

Every step shown. Box = final answer.

1. (2 pts)

- 1) a. $-\sqrt{-49} = -(i\sqrt{49}) = -7i$. So $a = 0$ and $b = -7 \neq 0$.
- 2) b. $6 - 2i$ has $a \neq 0$ and $b = -2 \neq 0$.
- 3) Every number is complex; $b \neq 0$ makes it nonreal complex; $a = 0$ with $b \neq 0$ makes it pure imaginary.

a. complex, pure imaginary, nonreal complex b. complex, nonreal complex

2. (2 pts)

- 1) Pull out i FIRST: $\sqrt{-8} = i\sqrt{8} = 2i\sqrt{2}$ and $\sqrt{-18} = i\sqrt{18} = 3i\sqrt{2}$.
- 2) Multiply: $(2i\sqrt{2})(3i\sqrt{2}) = 6i^2(\sqrt{2})^2 = 6i^2(2) = 12i^2$.
- 3) $i^2 = -1$, so $12(-1) = -12$. (Writing $\sqrt{144} = 12$ is the trap.)

-12

3. (2 pts)

- 1) a. $47 \div 4 = 11$ remainder 3, so $i^{47} = i^3 = -i$.
- 2) b. $14 \div 4 = 3$ remainder 2, so $i^{14} = i^2 = -1$.
- 3) $1 / (-1) = -1$.

a. -i b. -1

4. (2 pts)

- 1) Distribute the minus: $6 - 5i + 2 - 3i + 4 - i$.
- 2) Real parts: $6 + 2 + 4 = 12$.
- 3) Imaginary parts: $-5 - 3 - 1 = -9$.

12 - 9i

5. (2 pts)

- 1) FOIL: $(3)(-2) + (3)(5i) + (-4i)(-2) + (-4i)(5i) = -6 + 15i + 8i - 20i^2$.
- 2) $i^2 = -1$, so $-20i^2 = +20$.
- 3) $-6 + 20 + 23i$.

14 + 23i

6. (3 pts)

- 1) Multiply top and bottom by the conjugate of the bottom, $2 + 3i$.
- 2) Top: $(4 + 7i)(2 + 3i) = 8 + 12i + 14i + 21i^2 = 8 + 26i - 21 = -13 + 26i$.
- 3) Bottom: $(2 - 3i)(2 + 3i) = 4 + 9 = 13$.
- 4) $(-13 + 26i)/13 = -1 + 2i$.

-1 + 2i

7. (4 pts)

- 1) Set equal to 0: $2x^2 + 4x + 3 = 0$, so $a = 2$, $b = 4$, $c = 3$.
- 2) Discriminant: $b^2 - 4ac = 16 - 24 = -8$.
- 3) $x = (-4 \pm \sqrt{-8}) / 4 = (-4 \pm 2i\sqrt{2}) / 4$.
- 4) Split: $-4/4 = -1$ and $2\sqrt{2}/4 = \sqrt{2}/2$.

$x = -1 \pm (\sqrt{2}/2) i$

8. (3 pts)

- 1) Add real parts and imaginary parts: $(5 - 3) + (-2 + 6)i = 2 + 4i$.
- 2) Graph $5 - 2i$ as an arrow from $(0,0)$ to $(5, -2)$; graph $-3 + 6i$ as an arrow to $(-3, 6)$.
- 3) The resultant is the diagonal of the parallelogram they form: an arrow from $(0,0)$ to $(2, 4)$.

2 + 4i (resultant drawn to (2, 4))

9. (3 pts)

- 1) 300° is in quadrant IV, reference angle 60° .
- 2) $\cos 300^\circ = 1/2$, $\sin 300^\circ = -\sqrt{3}/2$.
- 3) $8(1/2) + 8(-\sqrt{3}/2)i = 4 - 4\sqrt{3}i$.

4 - 4i√3

10. (4 pts)

- 1) a. Sketch: $x = -3$, $y = -3$ is quadrant III.
- 2) $r = \sqrt{9 + 9} = \sqrt{18} = 3\sqrt{2}$.
- 3) $\tan \theta = (-3)/(-3) = 1$, reference angle 45° . Quadrant III: $\theta = 180^\circ + 45^\circ = 225^\circ$.
- 4) b. $-5 = -5 + 0i$ lies on the negative real axis: $r = 5$, $\theta = 180^\circ$.

a. $3\sqrt{2}(\cos 225^\circ + i \sin 225^\circ)$ b. $5(\cos 180^\circ + i \sin 180^\circ)$

11. (3 pts)

- 1) $z^2 - 1 = i^2 - 1 = -1 - 1 = -2$. $|-2| = 2$. Exactly 2 does NOT exceed 2, so keep going.
- 2) $(-2)^2 - 1 = 4 - 1 = 3$. $|3| = 3$.
- 3) $3 > 2$, so $z = i$ is not in the Julia set.

NO